Code No. 3528

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (AI&ML) V-Semester (MAIN) Examinations Dec 2023 /Jan 2024

ARTIFICIAL INTELLIGENCE

Time : 3 hours **Max. Marks : 70**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7).*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

		Marks	BT	CO
1. a)	Define rational agent.	2	1	1
b)	What is state space search.	2	1	1
c)	Distinguish between perfect and imperfect information games.	2	4	2
d)	Define skolemization.	2	2	2
e)	Obtain wwf representation of the statement: Any person who has an umbrella is not wet.	2	3	1
f)	Define uncertainty and list the reasons for uncertainty.	2	1	2
g)	List out any two applications of neural networks.	2	1	2
2. a)	Explain the structure of agent and various types of agents in block diagram for each.	7	2	2
b)	Write short notes on minmax procedure.	7	2	2
3. a)	Explain Alpha beta pruning with an example.	7	2	2
b)	Discuss about iterative deepening search algorithm with a diagram.	7	3	2
4. a)	Explain the semantic network of knowledge representation with an example in detail.	7	2	1
b)	What is resolution. Use propositional resolution to prove that: (~ U ^ S) can be proved from { A∨ C, ~ C =>B, ~B, A => S, ~U}	7	3	1

(P.T.O.)

5.	a)	Describe phases in building expert system and draw the architecture of expert system.	7	2	2
	b)	Write short notes on fuzzy inference system.	7	2	2
6.	a)	What is Symbolic AI. What are the limitations of Symbolic approach.	7	2	2
	b)	Describe Hopfield networks in detail.	7	1	2
7.	a)	Explain A* algorithm with an example and describe admissibility in A*	7	4	1
	b)	Determine whether the following formula is consistent or inconsistent using tableau method: $(\sim A \vee B) \wedge (C \rightarrow B) \wedge (A \vee C)$	7	5	1

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